Big Picture: CSCI 596 & Cyber-science Nexus

Aiichiro Nakano

Department of Computer Science
Department of Physics & Astronomy
Department of Quantitative & Computational Biology
University of Southern California

Email: anakano@usc.edu

Science at the nexus of post-Exaflop/s computing, quantum computing, and AI at USC

Changing Computing Landscape for Science

Postexascale Computing for Science

Deelman *et al.*,
Science **387**, 829 ('25)

Exaflop/s = 10^{18} mathematical operations per second

Compute Cambrian explosion

AI for Science

DOE readies multibillion-dollar AI push

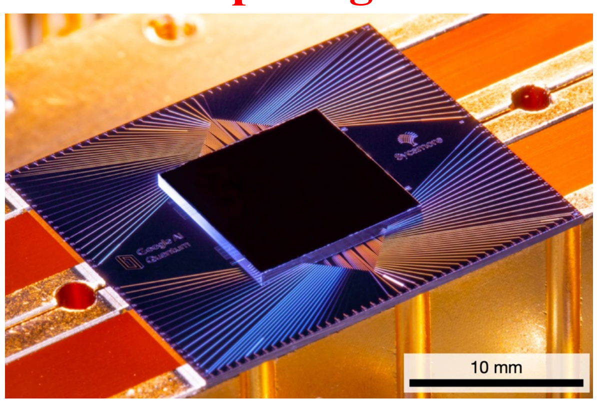
U.S. supercomputing leader is the latest big backer in a globally crowded field

By **Robert F. Service**, *in Washington, D.C.*

Science **366**, 559 ('19)

Quantum Computing for Science

Arute, Martinis *et al.*,
Nature **574**, 505 ('19)



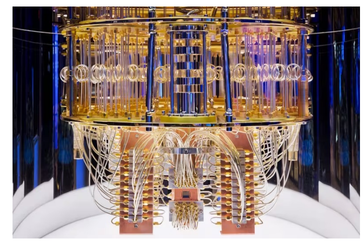
Google Sycamore

Use all to advance science & engineering!

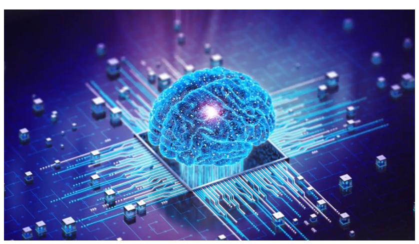
Training Cyber-Science Workforce

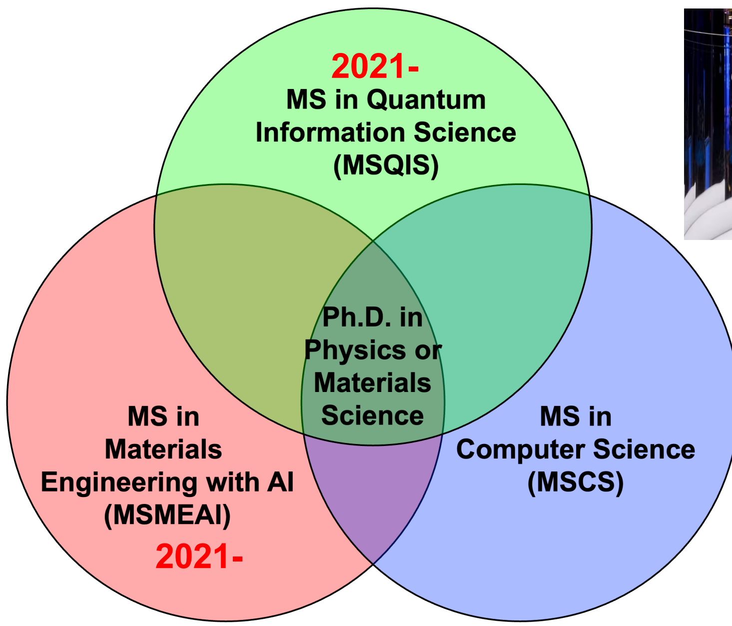
- New generation of computational scientists at the nexus of post-Exascale computing, quantum computing & AI
- Unique dual-degree program: Ph.D. in materials science or physics, along with MS in computer science, MS in quantum information science or MS in materials engineering with AI

Cybertraining on Exa + quantum + AI platforms


IBM Quantum

Neuromorphic Hala Point





Exascale Aurora

AIQ-XMaS Software Suite

AI & Quantum-Computing Enabled Exa Quantum Materials Simulator

PostExa

MLMD

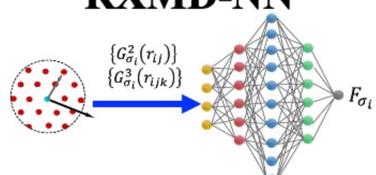
**Multiscal
e light-
matter
dynamics**

QXMD

Nonadiabatic quantum molecular dynamics

Aurora ESP engine

RXMD-NN



Reactive & neural-network molecular dynamics

GEARS

Augmented-reality user interface

EZFF

```
EZFF/
├── LICENSE
├── MANIFEST.in
├── README.md
├── docs
├── examples
├── ezff
│   ├── errors.py
│   ├── ffio.py
│   ├── interfaces
│   │   ├── gulp.py
│   │   ├── qchem.py
│   │   └── vasp.py
│   │   ├── lammps.py
│   │   └── rxmd.py
│   └── utils
│       ├── convert_units.py
│       └── reaxff.py
├── setup.py
└── tests
```

Easy force-field parameterization & uncertain quantification

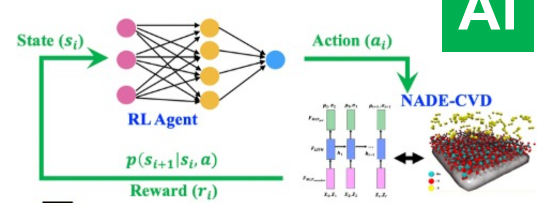
MISTIQS

Quantum

Quantum many-body dynamics on quantum computers

MAITAI

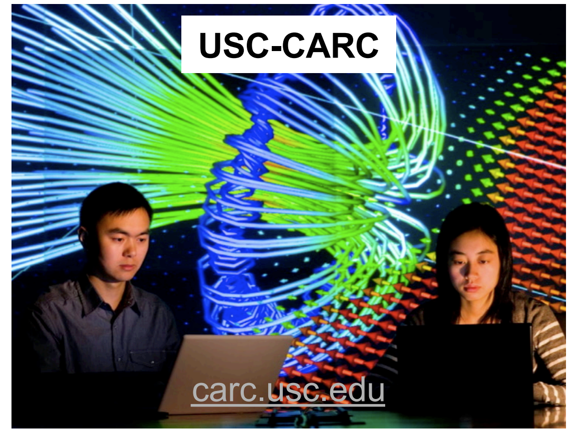
AI



AI tools for materials design

https://cybermagics.netlify.app/software

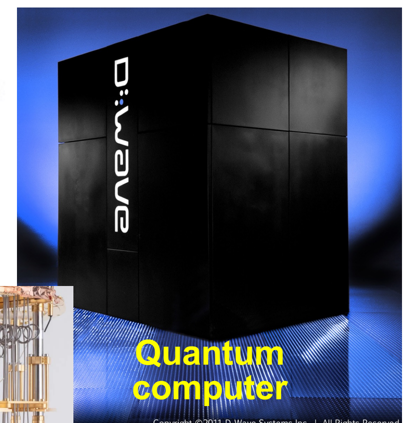
High-Performance Computing (HPC)



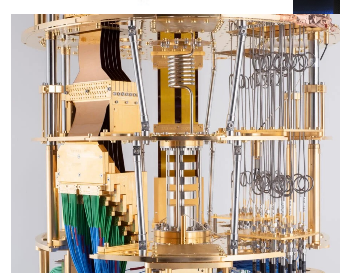
* Exaflop/s = 10^{18} mathematical operations per second

- *USC CARC (Center for Advanced Research Computing)*: **20,000 CPU cores & 200 GPUs**
- *USC ISI (Information Sciences Institute)*: **5,000+ qubit D-Wave quantum computer**
- *USC-IBM Quantum Innovation Center*: **Cloud access to 100+ qubit IBM quantum computers (2024-)**

GPU



QPU



Computational Sciences at USC

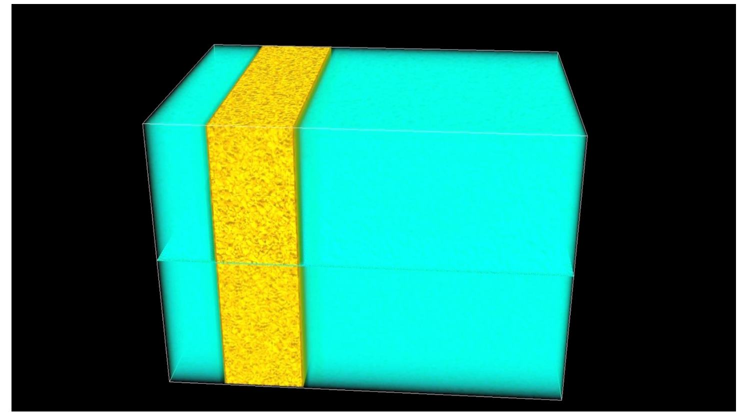
The Nobel Prize in Chemistry 2013

© Nobel Media AB
Martin Karplus

Photo: Keilana via Wikimedia Commons
Michael Levitt

Photo: Wikimedia Commons
Arieh Warshel

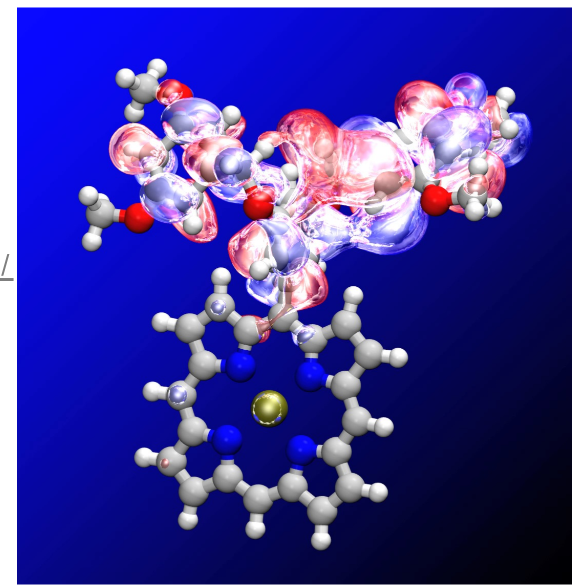
The Nobel Prize in Chemistry 2013 was awarded jointly to Martin Karplus, Michael Levitt and Arieh Warshel *"for the development of multiscale models for complex chemical systems"*.



Fields Medal 2026

Prof. Yu Dong (formerly at USC) resolved Hilbert's sixth problem: deriving fluid equations from Newton's laws by way of Boltzmann's kinetic theory.

https://dornsife.usc.edu/news/stories/yu-deng-wins-2026-fields-medal/

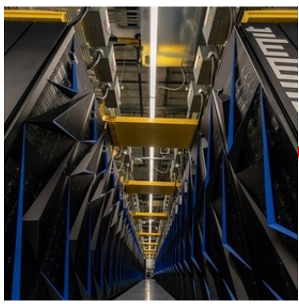


Collaboratory for Advanced Computing & Simulations cacs.usc.edu

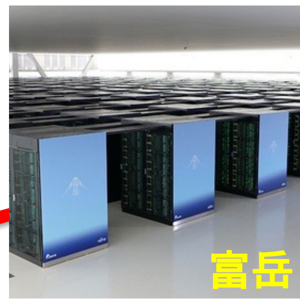
- **5.0 trillion-atom molecular dynamics**
- **40 trillion electronic degrees-of-freedom quantum molecular dynamics**
- **Million node-hrs/yr of computing on 21,248 CPUs + 63,744 GPUs**

Beyond Exaflop/s

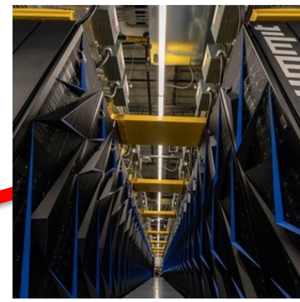
World's top1 supercomputer in recent years



Summit (ORNL, US)
Jun. 2018
0.15 Eflop/s
CPU/GPU



Fugaku (RIKEN, Jpn)
Jun. 2020
0.44 Eflop/s
CPU



Frontier (ORNL, US)
Jun. 2022
1.2 Eflop/s
CPU/GPU

El Capitan (LLNL, US)
Nov. 2024
1.8 Eflop/s
CPU/GPU

Lineshine (NSCS, Chn)
Jun. 2026
2.2 Eflop/s
CPU

Science **359**, 617 ('18)

Design for U.S. exascale computer takes shape

Competition with China accelerates plans for next great leap in supercomputing power

By **Robert F. Service**

In 1957, the launch of the Sputnik satellite vaulted the Soviet Union to the lead in the space race and galvanized the United States. U.S. supercomputer researchers are today facing their own Sputnik moment—this time with China. After dominating the supercomputing rank- Lemont, Illinois. That's 2 years earlier than planned. "It's a pretty exciting time," says Aiichiro Nakano, a physicist at the University of Southern California in Los Angeles who uses supercomputers to model materials made by layering stacks of atomic sheets like graphene.

Called A21, the Argonne computer will be built by Intel and Cray and is expected to

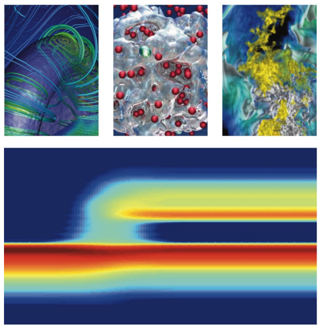
Exa-leadership

BES

BASIC ENERGY SCIENCES

EXASCALE REQUIREMENTS REVIEW

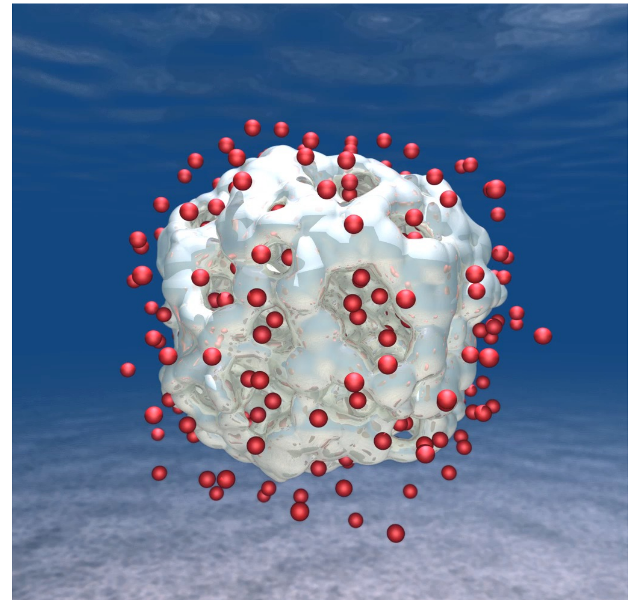
An Office of Science review sponsored jointly by
Advanced Scientific Computing Research and Basic Energy Sciences



NOVEMBER 3–5, 2015

ROCKVILLE, MARYLAND

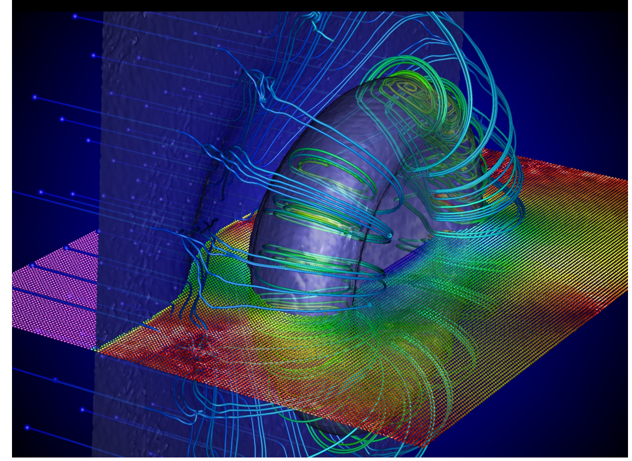
U.S. DEPARTMENT OF ENERGY



16,661-atom QMD

Shimamura *et al.*, *Nano Lett.* **14**, 4090 ('14)

On-demand hydrogen production from water

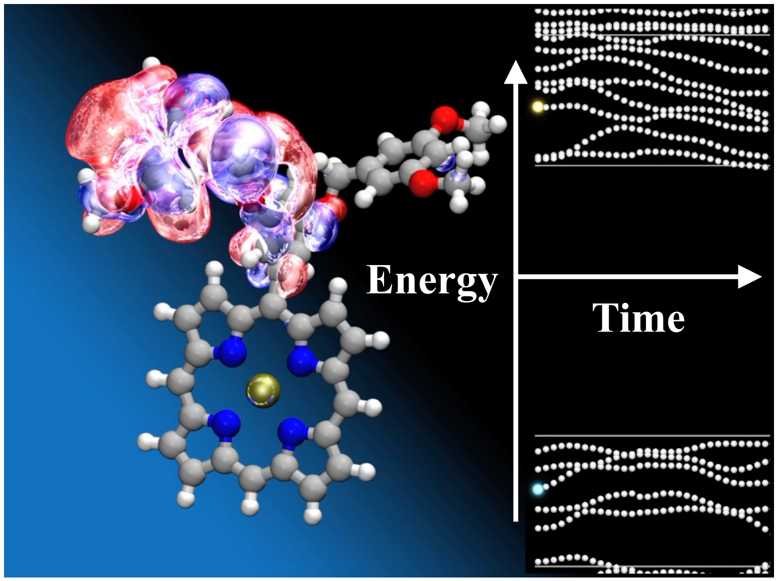


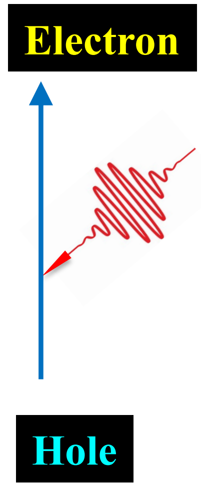
10^9-atom RMD

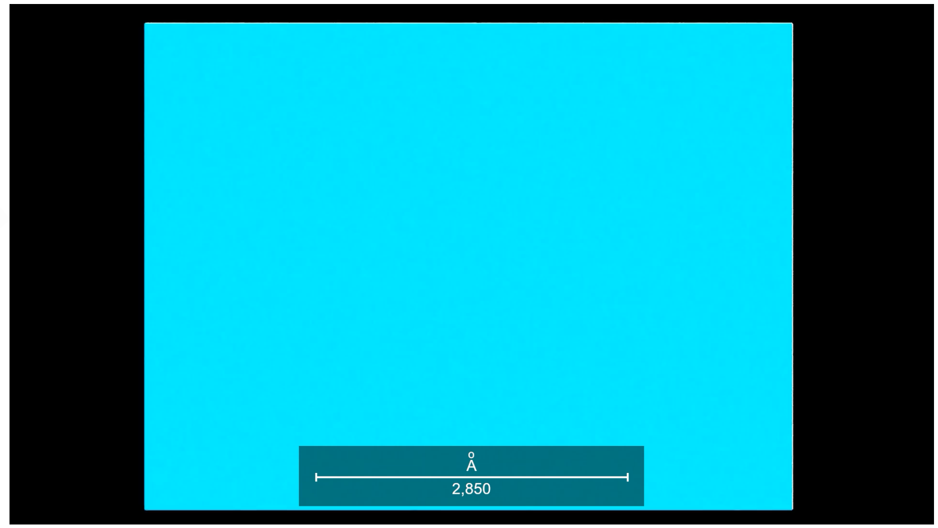
Shekhar *et al.*, *Phys. Rev. Lett.* **111**, 184503 ('13)

Fluid dynamics atom-by-atom

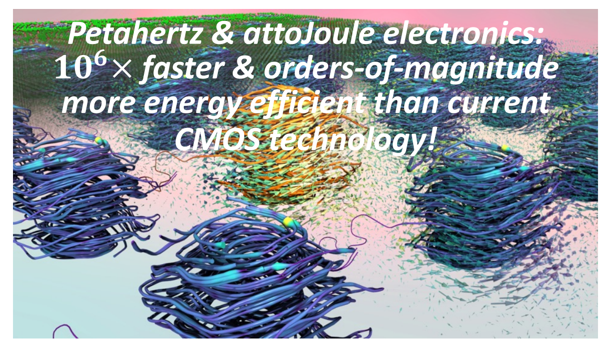
Multi-Exaflop/s Quantum Mechanics







That's what we do



Argonne's Aurora supercomputer drives groundbreaking simulations to explore how light shapes quantum materials

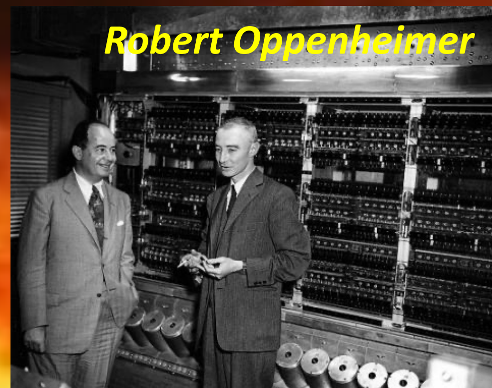
A USC-led team uses exascale computing power and AI to simulate light-matter interactions at an unprecedented scale

The team's work has been named a finalist for the Association of Computing Machinery's Gordon Bell Prize, one of the top honors in high-performance computing.

T. M. Razakh *et al.*, *Proc. SC* ('25) doi: 10.1145/3712285.3771785

In the Beginning …

John von Neumann



Manhattan project

Los Alamos, NM, 1945

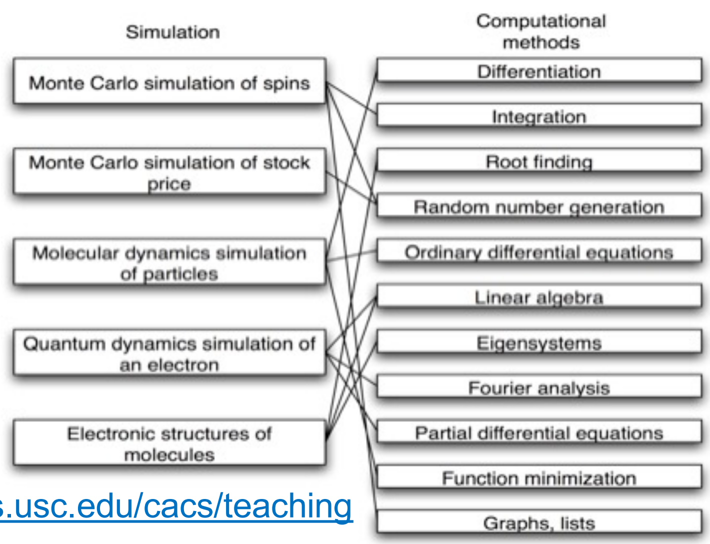
Journey of Electrons and Atoms: Personal Story of Quantum Mechanics & Supercomputing (Nov. 16 & 17, '24, Culver City, CA)

https://magazine.viterbi.usc.edu/spring-2025/engineering/just-dance/

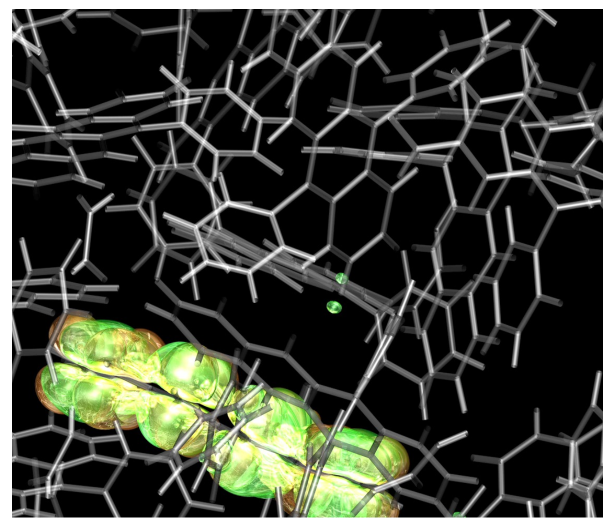
CACS HPCS Courses

- **CS596: Scientific Computing & Visualization**
  Hands-on training on particle/field simulations, parallel computing, & scientific visualization (MPI, OpenMP, CUDA, OpenGL)
- **CS653: High Performance Computing & Simulations**

  Deterministic/stochastic simulations, scalable parallel/Grid computing, & scientific data visualization/mining in virtual environment
- **Phys516: Methods of Computational Physics**
  Numerical methods in the context of physics simulations



https://sites.usc.edu/cacs/teaching



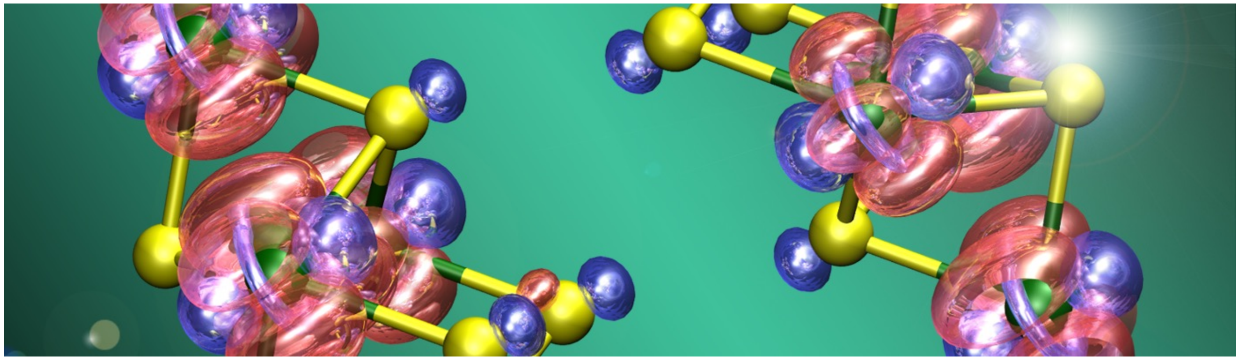
Additional HPCS Course

Detailed lecture notes are available at a USC course home page

PHYS 760: EXTREME-SCALE QUANTUM SIMULATIONS

Course Description

Computer simulation of quantum-mechanical dynamics has become an essential enabling technology for physical, chemical & biological sciences & engineering. Quantum-dynamics simulations on extreme-scale parallel supercomputers would provide unprecedented predictive power but pose enormous challenges as well. This course surveys & projects algorithmic & computing technologies that will make quantum-dynamics simulations metascalable, *i.e.*, "design once, continue to scale on future computer architectures".



https://aiichironakano.github.io/phys760.html

Related Course

CSCI 452 (EE 451): Parallel Programming
Prof. Viktor Prasanna

Topics: Parallel computation models, message passing & shared memory paradigms, data parallel programing, performance modeling & optimization, memory system optimization techniques, fine grained computation models & high-level design tools for programming parallel platforms, communication primitives, stream programming models, emerging heterogeneous computing & programming models.

This course will study the abstractions for parallel programming as well as provide students with hands-on experience with state-of-the-art parallel computing platforms & tools including large scale clusters, edge devices & data center scale platforms.

https://aiichironakano.github.io/cs596/EE451.pdf

- **To replace former CSCI 503 (Parallel Programming)**

CARC Tutorials & Office Hours

Series of tutorials + office hours (T, 2:30-5 pm, Leavey Library 3L) by USC Center for Advanced Research Computing (CARC):

- **Introduction to Python, R**
- **Parallel MATLAB**
- **...**

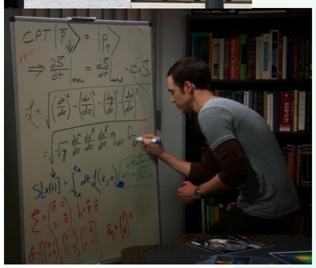
https://www.carc.usc.edu/education/workshops
https://www.carc.usc.edu/user-support/office-hours-and-consultations

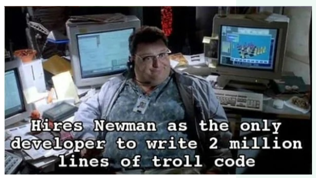
Students registered by the end of this week will get a CARC account

MS in Quantum Information Science

- **MS degree in Quantum Information Science (MSQIS) started in 2021**
- **Required foundational courses**
  1. **EE 520: Introduction to Quantum Information Processing**
  2. **EE 514: Quantum Error Correction**
  3. **Phys 513: Applications of Quantum Computing**
- **Core — at least two courses from**
  1. **EE 589: Quantum Information Theory**
  2. **Phys 550: Open Quantum Systems**
  3. **Phys 559: Quantum Devices**
  4. **Phys 660: Quantum Information Science & Many-Body Physics**
- **Phys 513: Application of Quantum Computing (co-taught with Prof. Rosa Di Felice) — quantum simulations on quantum circuits & adiabatic quantum annealer (syllabus)**
- **Coming soon: Progressive (4+1) degree with BS in CS, EE, math, physics, *etc*.**

New: Life of a Computational Scientist





Repeat:
Random thought (scientific problem?)
↓ Reconnection of semantic network
Work out math
↓
Operationally understand
↓ What that means in your words? (Richard Feynman)
Write program
↓
Run program
↓
Scientific finding (positive or negative)
↓
So what? Now what?

Now with Agentic AI

Large language model (LLM) plan

Symbolic math: https://openai.com/index/harness-engineering/

SimPy www.sympy.org

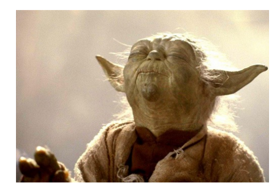
Common internal representation

Coding agents:
OpenAI Codex, Claude Code
chatgpt.com claude.com/product/claude-code

Agentic skills

https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills

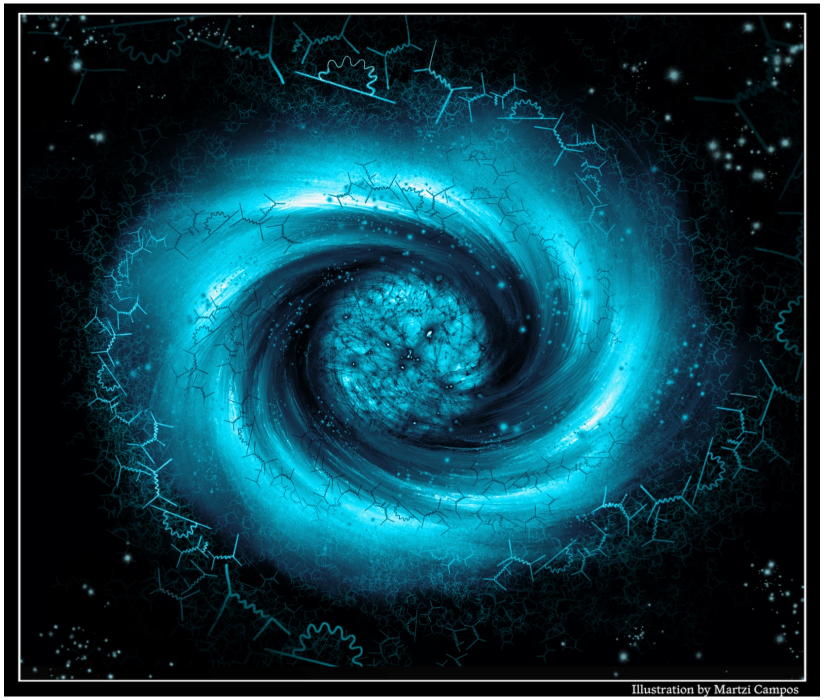
Then → **Now**



AI is here—use it: “There is no going back” (Matthew Schwartz)
… starting with CSCI 596

Now What?

- What does it mean to "understand" at the age of AI-assisted (black-box) science?



Lyman Alpha forest Research Collaboration (LARC)
PIs—Vera Gluscevic (*USC*)
$4M John Templeton Foundation award
(Dec. 2025 – Nov. 2028)
to *USC*, *UC Riverside & Carnegie Observatories*

- Cosmology (Gluscevic, Bird, *et al*.), computer science (Shelton, Nakano), cinematic arts & philosophy (Gallow) collaboration on "Growing a universe from quantum scales: discovery at the intersection of cosmology, *epistemology* & interactive visual media"

https://dornsife.usc.edu/news/stories/4m-templeton-foundation-grant-funds-new-frontier-in-cosmology/